Internet of Things: Protocols and System Software

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Índice general

1. Introduction

In the current technological world, there are three different main trends:

1. Computing: computers are getting smaller, cheaper and more powerful.

2. Sensing: there is a huge variety of sensors that can be used to measure different physical properties. They are also getting smaller and cheaper.

3. Transmission: two factors are being taken into account:

   - There are more and more wirelesscommunication technologies, with different characteristics. This leads to omnipresent connectivity.
   - Energy consumption is a key factor in the design of these technologies, as many IoT devices are battery-powered. This is leading to the development of low-power communication protocols and energy-efficient hardware.

In this context, the Internet of Things (IoT) is emerging as a paradigm where everyday objects become smart and connected to the internet.

Definición 1.1 (IoT Thing)**.** An IoT thing is a physical object that is augmented with embedded electronics, making it therefore “smart”.

1.1. Digital Twin

The Digital Twin is a global representation of the world, where every physical object has a digital counterpart. This means that every physical object is represented in the digital world, and every change in the physical world is reflected in the digital world and vice versa. This is a very ambitious vision, and it is still far from being achieved. However, it is a useful concept to understand the potential of IoT and the challenges that it poses. Its three main features are:

- Live: the digital twin is continuously updated with data from the physical world.

- Predictive: the digital twin can be used to predict the behavior of the physical world using simulations and machine learning techniques.

- Interactive: depending on the predictions made by the digital twin, it can change and therefore make changes in the physical world.

In this situation, the world would become observable and programmable in real time, which would have a huge impact on many different fields, such as healthcare, transportation, manufacturing, etc. However, there are still many challenges that need to be overcome to achieve this vision.

1.2. IoT Views

There are two different views on IoT:

- IoT as a network of connected *things*, aka *M2M (Machine to Machine)*.

  It is quite a low-level technical view, where the focus is on the devices and the communication between them. Internet is just seen as a communication medium, and the main goal is to connect as many things as possible.

  It is the view that is usually taken by engineers and computer scientists.

- IoT as a network of connected *data*, aka *Big Data*.

  It is a more high-level view, where there are still things but the focus is on the data that is generated by these things and how it can be used to create value. Internet is seen as a platform for data collection and analysis, and the main goal is to extract insights from the data.

  It is the view that is usually taken by business people and data scientists.

2. IoT Systems

An IoT system consists of all the HW and SW components that are required to implement an IoT application. In order to reduce the complexity of an IoT system, developpers tend to reuse the same components across different applications, and here is where the concept of *IoT platforms* comes into play.

2.1. IoT Platforms

An IoT platform is a framework that provides a set of tools and services to facilitate the development, deployment, and management of IoT applications. By using an IoT platform, developers can focus on the application logic rather than dealing with the complexities of the underlying infrastructure. There are many IoT platforms available in the market, each with its own strengths and weaknesses. There are two main categories of IoT platforms:

- <u>Horizontal</u>: These are general-purpose platforms that can be used for a wide range of IoT applications. They provide a broad set of features and services that can be customized to fit the specific needs of different applications.

- <u>Vertical</u>: These platforms are designed for specific industries or use cases. They provide specialized features and services that are tailored to the requirements of a particular domain.

2.2. IoT System Architecture

An IoT system architecture defines whole IoT system. Regarding the HW, it consists of 4 different types of components:

1. <u>IoT Things</u>: Also known as Smart Objects, these are the physical devices that actually interact with the environment.

2. <u>Gateways</u>: These are devices that act as intermediaries between the IoT Things and the Internet. Given that the IoT Things are often resource-constrained and may not have the capability to connect directly to the Internet, IoT Gateways provide the necessary connectivity and protocol translation.

3. <u>Internet</u>: This is the communication infrastructure that allows the IoT Gateways to connect and exchange data with other devices and services.

4. Online Services: These are cloud-based platforms and services that provide different functionalities to support the IoT applications.

However, SW components of an IoT system are also crucial to the overall functionality and performance of the system. There are three main aspects from the SW perspective that affect the whole IoT system.

2.2.1. SW Functional Blocks

From the SW perspective, it is essential to identify the functional blocks that are required to implement an IoT application. It is difficult to define a standard set of functional blocks that can be applied to all IoT applications, but for the aim of this course, we will consider three functional blocks:

1. IoT Networking: SW necessary to enable communication between all the components of the IoT system.

2. IoT Data Management: SW responsible for measuring, collecting, storing, and making data available for querying and analysis.

3. IoT Data Processing: SW responsible for analyzing the data collected from the IoT Things and extracting useful insights and information.

   There are two main recipient groups of the data in order to make use of it:

   - Humans: The data is presented to the users in a human-readable format, such as dashboards, reports, or visualizations.
   - Algorithms: The data is used as input for machine learning models, predictive analytics, or other algorithms that can make decisions or take actions based on the data.

   There are four main levels of data processing that can be applied to the data collected from the IoT Things:

   a) Descriptive: The data is only used to observe and describe the current state of the system.

   b) Diagnostic: The data is used to identify the causes of certain events or conditions in the system and explain why they happened.

   c) Predictive: The data is used to anticipate future events or conditions in the system based on historical data and patterns.

   d) Prescriptive: The data is used to perform actions or make decisions based on the insights derived from the data analysis. This has several legal and ethical implications, and usually requires human supervision and intervention to ensure that the actions taken are appropriate and that someone is legally responsible for the consequences of those actions.

2.2.2. SW Blocks Placement

The SW functional blocks can be executed in different specific HW components within the IoT system, depending on the specific requirements and constraints of the application. There are three main approaches for the placement of the SW functional blocks.

1. Cloud-based IoT Systems: The "classic" approach.

   In this approach, the biggest part of the three SW functional blocks are executed in the cloud.

   - IoT Things are responsible for collecting data and sending it to the gateway. Therefore, they only execute a small part of the IoT Networking and IoT Data Management blocks.
   - IoT Gateways are responsible for receiving the data from the IoT Things and forwarding it to the cloud. Therefore, they execute a small part of the IoT Networking.
   - The online services in the cloud are responsible for executing the majority of the three SW functional blocks, including IoT Networking, IoT Data Management, and IoT Data Processing.

2. Edge-based IoT Systems: The "modern" approach.

   This approach is similar to the cloud-based approach, but with a significant shift of the SW functional blocks towards the gateways. Given that the online services have limited resources and are under high load, it is necessary to offload some of the processing tasks to the gateways in order to reduce latency and improve performance.

3. Mesh-based IoT Systems: The "future" approach.

   In this approach, the SW functional blocks are even for shifted towards the IoT Things, which are becoming more powerful and capable of executing more complex tasks. This approach just considers gateways and online services as fallback options for the IoT Things in case they are not able to execute certain tasks.

2.2.3. SW Blocks Communication

The SW functional blocks need to communicate with each other in order to exchange data and coordinate their actions. This communication can be achieved through different connectivity options and protocols, depending on the specific requirements and constraints of the application. We will explore them in more detail in the rest of the course.

3. IoT Internetworking

During this chapter, we will focus on the IoT internetworking aspect, which is crucial for enabling communication and data exchange between different components of an IoT system. A reader may initially think that using the Internet and the well-known TCP/IP protocol suite would be sufficient for enabling communication in an IoT system, and it is in fact a good approach as these protocols have proven to be extremely successful and widely adopted in the traditional Internet. However, there are several challenges and limitations that need to be addressed when using the Internet and TCP/IP protocols in the context of IoT systems, as an usuall Internet device is very different from an IoT device in terms of resources, capabilities, and requirements. Some of these challenges and limitations include:

- Data Item Size: IoT devices often generate and exchange small data items, such as sensor readings or control commands, while the Internet and TCP/IP protocols are designed to handle larger data items, such as web pages or multimedia content.

- Not Optimized for Energy Efficiency: IoT devices are often battery-powered and have limited energy resources, while the Internet and TCP/IP protocols are not optimized for energy efficiency and may require more power consumption than what is feasible for IoT devices.

- Large Buffer Requirements: The Internet and TCP/IP protocols may require larger buffer sizes than what is available on IoT devices, which can lead to performance issues and increased latency.

- Pull vs. Push Communication: The Internet and TCP/IP protocols are primarily designed for pull-based communication, where clients request data from servers, while IoT systems often require push-based communication, where devices need to send data to other devices or applications without waiting for a request.

In order to address these challenges and limitations, there are several IoT-specific protocols and technologies that have been developed to enable efficient and effective communication in IoT systems. There are two main approaches to connect the IoT Things to the Internet:

- Application layer proxy:

  There is a proxy (usually implemented in the IoT gateway) that translates between the protocols used by the IoT Things and the protocols used by the Internet.

- The IoT Thing communicates with the proxy using dedicated IoT protocols that have nothing to do with the Internet protocols.
- The proxy stores the data received from the IoT Things and makes it available to the Internet using standard Internet protocols.

An example of this approach is when an user has to install a specific app on their smartphone in order to interact with a smart home device, such as a smart thermostat or a smart light bulb.

- IP Router:

  In this approach, the IoT Things directly use IP, but with some optimizations and adaptations to address the challenges and limitations mentioned above. An usual router may be combined (or may replace) with the IoT gateway to enable the IoT Things to connect to the Internet using IP. That way, the IoT Things can communicate directly with other devices and applications on the Internet without the need for a proxy, which is much more scalable. There is no need for specific apps or software to interact with the IoT Things, as they can be accessed using standard Internet protocols and interfaces.

During the rest of the course, we will explore in more detail the second approach, which is the one that is becoming more and more popular in the IoT ecosystem, as it provides better scalability and interoperability.

3.1. IEEE 802.15.4

IEEE 802.15.4 is a standard for low-rate wireless personal area networks (LR-WPANs) that is widely used in IoT applications. Its main focus is on providing low-power, low-cost and low-complexity wireless communication for IoT devices. It defines the Physical Layer (PHY) and part of the Data Link Layer (MAC). In addition, it defines multiple variants for each of the layers, which can be used in different scenarios and applications depending on the specific requirements and constraints of the IoT system.

Communication using this standard is based on a IEEE 802.15.4 PAN, which is a collection of devices that are connected together using the IEEE 802.15.4 standard. The members of a PAN can either be:

- Full Function Devices (FFDs): These devices can communicate with any other device in the PAN, including other FFDs and Reduced Function Devices (RFDs). They can also take over network management activities.

  If two devices need to communicate with each other but they are not in range of each other, they can use an FFD to forward messages between them, which allows for multi-hop communication and extends the range of the PAN. Only FFD have this capability.

- Reduced Function Devices (RFDs): These resource-constrained devices can only communicate with one FFD in the PAN, and they cannot take over network management activities.

In addition, each of the members of a PAN can have different roles, which determine their behavior and responsibilities in the network:

- PAN Coordinator: This is the device that is responsible for managing the PAN, including tasks such as the PAN creation (by selecting a PAN ID and a channel), adding and removing devices from the PAN, and coordinating the communication between devices in the PAN. There can only be one PAN Coordinator in a PAN, and it must be an FFD.

- Coordinator: This is a device that is responsible for managing a portion of the PAN, such as a cluster of devices. It can also take over some of the network management activities, such as routing and forwarding of messages. There can be multiple Coordinators in a PAN, and they must be FFDs.

- Members: These are the devices that are part of the PAN and can communicate with other devices in the PAN. They can be either FFDs or RFDs, and they do not have any specific responsibilities or roles in the network.

Once the members of a PAN have been defined and assigned their roles, they can communicate with each other using one of these two types of topologies:

- Star Topology: In this topology, all the devices in the PAN communicate directly with the PAN Coordinator, which acts as a central hub for the communication. This topology is simple and easy to manage, but it can be less efficient and less scalable than other topologies, as all the communication has to go through the PAN Coordinator, which can become a bottleneck and a single point of failure.

- Peer-to-Peer Topology: In this topology, the devices in the PAN can communicate directly with each other without the need for a central hub. This topology is more efficient and scalable than the star topology, as it allows for direct communication between devices, which can reduce latency and improve performance. However, it can be more complex to manage, as it requires more coordination and synchronization between devices to ensure that messages are delivered correctly and efficiently.

It should however be noted that the IEEE 802.15.4 standard does not define a protocol but only a standard for the physical and data link layers, so specific protocols as Zigbee have been developed on top of it to provide all the necessary functionalities and features for enabling communication in IoT systems.

3.1.1. Physical Layer (PHY)

The Physical Layer (PHY) of the IEEE 802.15.4 standard defines the physical characteristics and parameters of the wireless communication, such as the frequency bands where the communication can take place. It defines three different frequency bands that can be used for communication:

- 868 MHz Band: This band is used in Europe and other regions, and it provides a data rate of up to 20 kbps. It has one channel available for communication.

- 915 MHz Band: This band is used in North America and other regions, and it provides a data rate of up to 40 kbps. It has ten channels available for communication.

- 2.4 GHz Band: This band is used worldwide, and it provides a data rate of up to 250 kbps. It has sixteen channels available for communication.

The services provided by the PHY layer are divided into two categories:

- Data Services: These services are used for transmitting and receiving the PPDU between devices in the PAN.

- Management Services: These services are used for managing the communication and the network, such as channel scanning (to find the best channel for communication), link quality indication (to measure the quality of the communication link between devices) and CCA (Clear Channel Assessment, to check if the channel is idle before transmitting).

The content that is in fact transmitted over the medium is the *PHY Protocol Data Unit (PPDU)*, which consists of the *PHY Service Data Unit (PSDU)* encapsulated with the header. The 6 bytes Header consists of:

- Synchronization Header (SHR), 5 bytes:

  This header is used for synchronization between the transmitter and the receiver, and it consists of a preamble (4 bytes) and a start of frame delimiter (1 byte).

- PHY Header (PHR), 1 byte:

  The first 7 bits of this header are used to indicate the length of the PSDU, while the last bit is reserved for future use. Therefore, the maximum length of the PSDU is $2^7 - 1 = 127$ bytes.

3.1.2. Data Link Layer (MAC)

The standard IEEE 802.15.4 only defines the MAC sublayer of the Data Link Layer, which is responsible for managing access to the medium and ensuring reliable communication between devices in the PAN. The services provided by the MAC layer are divided into two categories:

- Data Services: These services are used for transmitting and receiving the MPDU between devices in the PAN across the PHY layer.

- Management Services: These services are used for managing the communication and the network, such as the beacon or GTS management. It also has hooks[1] for implementing security services.

[1]It has hooks (and not specific services) because the security services are eventually broken, so the standard would end up being unsecure nd obsolete. However, providing hooks allows for selecting the optimal security services and replacing them when they are broken without the need to change the standard itself.

The content that is in fact transmitted over the medium is the *MAC Protocol Data Unit (MPDU)*, which consists of the *MAC Service Data Unit (MSDU)* encapsulated with the header and the footer. The footer is just a 2 bytes CRC (Cyclic Redundancy Check) for error detection, while the header consists of:

- Frame Control (FC), 2 bytes:

  This field is used to indicate the type of frame being transmitted, such as a data frame, a beacon frame, or an acknowledgment frame.

- Sequence Number, 1 byte:

  This field is used to identify the frames being transmitted, which can be useful for ensuring reliable communication and detecting duplicate frames.

- Addressing Fields, 0 to 20 bytes:

  These fields are used to indicate the source and destination addresses of the frame being transmitted, and the PAN ID (2 bytes) of each device.

Regarding the MAC adresses, there are two types of addresses that can be used in the IEEE 802.15.4 standard:

- Short Address: This is a 16-bit address that is manually assigned to a device by the network operator. It is used for communication within the PAN, and it is not globally unique.
- Extended Address: This is a 64-bit address that is asigned to a device by the manufacturer, and it is globally unique. It is used for communication between devices in different PANs, and it can also be used for communication with devices on the Internet.

This standard also defines several types of variants for the MAC layer, which esentially use different variants of controlling the access to the medium.

CSMA/CA and variants

The Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) is a protocol for controlling access to the medium in a shared communication channel. It is based on the principle of carrier sensing, where devices listen to the channel before transmitting to ensure that it is not already in use by another device. Its main idea is described in Figure 3.1, where a backoff time is waited because:

- If multiple devices want to transmit at the same time and no backoff time is used, they would all start transmitting simultaneously when they sense that the channel is idle, which would lead to collisions and failed transmissions.
- It saves energy, as the device can go to sleep during the backoff time and wake up just before attempting to transmit, which can be more efficient than continuously sensing the channel.
- It allows for better performance, as it allows for the ackoledgment of successful transmissions and the retransmission of failed transmissions, which can improve the overall reliability and efficiency of the communication.

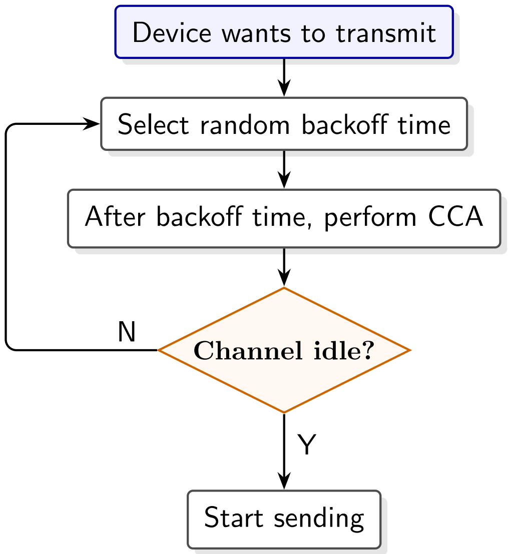

Figura 3.1: CSMA/CA protocol.

However, this main protocol has different specific variants that are used in the different variants of the MAC layer defined in the IEEE 802.15.4 standard.

- Slotted CSMA/CA:

  This variant is used when the frame is divided into slots. The main idea of this protocol is described in Figure 3.2. Some important abbreviations used there are:

  - *BE*: backoff exponent, which is used to calculate the possible number of backoff periods[2] that the device can choose for waiting.
  - *NB*: number of backoffs, which is the number of times the device has attempted to access the channel and has been unsuccessful.
  - *CW*: contention window, which is the number of consecutive clear channel assessments (CCAs) that the device needs to perform before it can start transmitting. It is used to ensure that the device has a good chance of successfully accessing the channel before it starts transmitting.

- Unslotted CSMA/CA:

  This variant is used when the time is not divided into slots. The main idea of this protocol is described in Figure 3.3. Here, the CW is not used as there are no slots, so the device can attempt to transmit as soon as it senses that the channel is idle, without waiting for the next slot.

- Time Division Multiple Access (TDMA):

  This is another variant used for controlling access to the medium when more than one device needs to transmit at the same time. In this protocol, the devices are assigned the whole frequency band for a specific period of time, which is called a time slot.

[2] A backoff period is the basic unit of time when using this protocol.

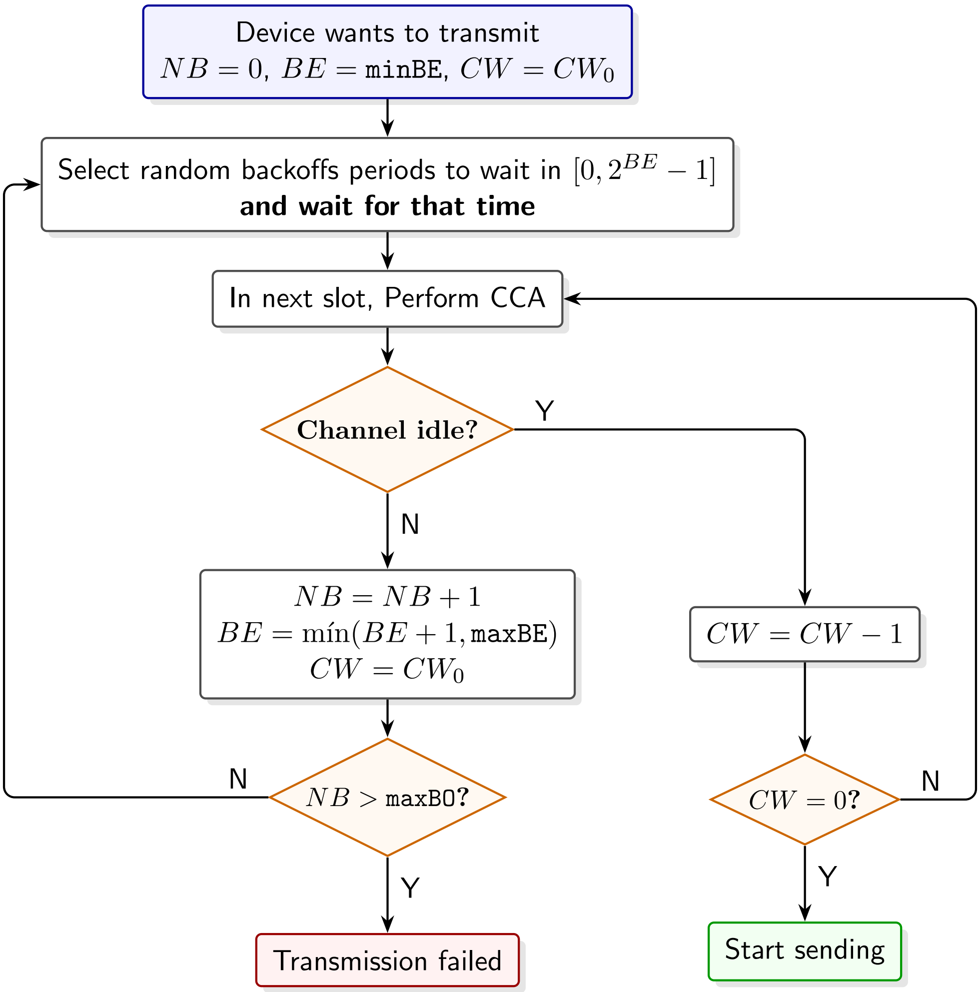

Figura 3.2: Slotted CSMA/CA protocol.

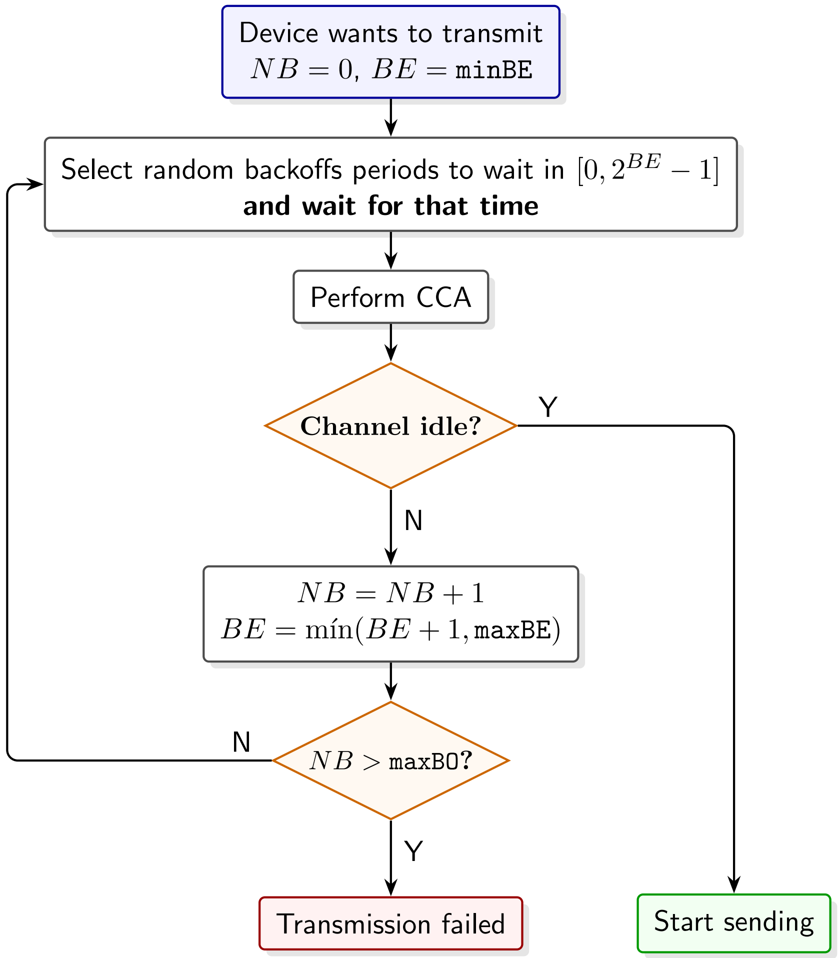

Figura 3.3: Unslotted CSMA/CA protocol.

- Frequency Division Multiple Access (FDMA):

  This is another variant used for controlling access to the medium when more than one device needs to transmit at the same time. In this protocol, the devices are assigned different frequency bands for communication, which allows them to transmit simultaneously without interference.

- Time-Frequency Division Multiple Access (TFDMA):

  This last variant is a combination of TDMA and FDMA, where the devices are assigned only a portion of the frequency band for a specific period of time, which allows for more efficient use of the available resources and can improve the performance of the communication.

Once the different forms of controlling access to the medium have been described, the specific variants of the MAC layer defined in the IEEE 802.15.4 standard can be described, which basically use different variants of controlling access to the medium.

Beacon-enabled Networks

Beacon-enabled networks are a variant of the MAC layer defined in the IEEE 802.15.4 standard, where time is divided into superframes. Each superframe starts with a *beacon frame* that is transmitted by the PAN Coordinator to synchronize the devices in the PAN and to provide information about the network, such as the PAN ID, the channel, and the superframe structure. It is divided in:

- Active Period: During this period of time, the devices in the PAN can communicate with each other. The active period is divided into 16 slots, and they are split into two parts (configurable by the PAN Coordinator):
  - *Contention Access Period (CAP)*: During these time slots, the devices in the PAN compete for access to the medium using the slotted CSMA/CA protocol to transmit their data. It should be noted than during the CAP, devices will only start transmitting if they have enough time slots (CW) to complete the transmission of their data.
  - *Contention Free Period (CFP)*: These time slots are called *Guaranteed Time Slots (GTS)*, and there are up to 7 GTSs available in each superframe. Devices in the PAN can request GTSs from the PAN Coordinator, which assigns them to the devices that request them. During these time slots, the devices that have been assigned GTSs can transmit their data without competing for access to the medium.
- Inactive Period: During this period of time (configurable by the PAN Coordinator), no communication takes place, and the devices in the PAN (including the PAN Coordinator) can go to sleep to save energy.

In this variant, there is also an Acknowledgment mechanism. In the header of a data frame sent, there is a specific bit that indicates whether the sender of the frame wants to receive an acknowledgment from the receiver (except from beacon and acknowledgment frames, which do not allow for acknowledgment). If this bit is set:

- If the receiver successfully receives the frame, it sends an acknowledgment frame back to the sender to confirm that the frame has been received correctly.
- If the sender does not receive an acknowledgment frame within a certain period of time, it assumes that the transmission has failed and attempts to retransmit the frame. There is a maximum number of retransmissions that the sender can attempt before giving up and considering the transmission as failed.

Networks without Beacons

This is another variant of the MAC layer defined in the IEEE 802.15.4 standard, where there are no beacon frames and no superframes. In this variant, the devices in the PAN continuously compete for access to the medium using the unslotted CSMA/CA protocol to transmit their data.

Time Slotted Channel Hopping (TSCH) MAC Layer

This new modern variant of the MAC layer defined in the IEEE 802.15.4 standard is designed to provide high reliability and low latency communication for IoT applications. The PAN coordinator divides time into slots, which are grouped into *slotframes*. Each slotframe consists of a fixed number of time slots, and the communication between devices in the PAN is organized around these slotframes. In addition, different frequency channels are used for communication, so a TFDMA approach is used for controlling access to the medium. Each time slot for each frequency channel is called a *cell*, and there are three types of cells:

- <u>Shared Cells</u>: These cells are used for communication between devices in the PAN, and they are shared among all the devices in the PAN. They use CSMA/CA for controlling access to the medium.
- <u>Dedicated Cells</u>: These cells are assigned to specific devices in the PAN, and they are used for communication between those devices. They are assigned using a pseudo-random algorithm[3] (it is not totally random because it needs to be deterministic to ensure that the devices know which cells they can use for communication), and they do not use CSMA/CA for controlling access to the medium, as the devices that have been assigned those cells can transmit their data without competing for access to the medium.
- <u>Advertisement Cells</u>: These cells are used for transmitting frames from the PAN Coordinator to the devices in the PAN.

It should be noted that this variant also uses an acknowledgment mechanism similar to the one used in the beacon-enabled networks, and both the data and the acknowledgment frames are supposed to be transmitted within the same cell.

For each slotframe, and given that there are multiple channels (frequencies) available for communication, the PAN Coordinator assigns a specific channel to each time slot in the slotframe, which is called *channel hopping*.

[3]IEEE 802.15.4 does not specify this. It is delegated to higher layer protocols.

3.2. NB-IoT

Even though the IEEE 802.15.4 standard and its variants are widely used in IoT applications, they have a limited range, which can be a problem for some IoT applications that require long-range communication. To address this issue, the Narrowband Internet of Things (NB-IoT) standard was developed to provide a low-power, wide-area (LPWA) communication solution for IoT applications. It is based on the Long-Term Evolution (LTE) standard, which is widely used for cellular communication.

3.2.1. LTE and NB-IoT

LTE (Long-Term Evolution) is a standard for wireless communication that is widely used for cellular communication over a licensed spectrum. It divides the whole area in smaller regions called *cells*, and each cell is served by a base station that provides wireless communication to the devices in that cell. It has a layered architecture that consists of three main layers:

- Host-to-Network Layer: This is the part of the LTE network that provides wireless communication to the devices in the cells. Each cell is served by a base station (called `eNB`, evolved Node B) that provides wireless communication to the devices (called `UE`, User Equipment) in that cell. The cellular network is responsible for managing the communication between the devices and the base stations.

- Core Network Layer: This is the part of the LTE network that provides the necessary infrastructure and services for enabling communication between the different cells and the Internet. They provide most of the functionalities and features of the LTE network, such as routing, mobility management, security, and quality of service (QoS). In this layer, multiple servers are used:

  - *Mobility Management Entity (MME)*: This server is responsible for managing the mobility of the devices in the LTE network, such as tracking their location and handling handovers between cells when they move.
  - *Home Subscriber Server (HSS)*: This server is responsible for storing the subscriber information and providing authentication and authorization services for the devices in the LTE network. Since LTE works in a licensed spectrum, only authorized devices can connect to the network, and the HSS is responsible for ensuring that.
  - *Serving Gateway (SGW)*: This server is responsible for routing the data packets between the devices in the LTE network and the Internet, and it also provides some security and QoS features.
  - *Packet Data Network Gateway (PGW)*: This server is responsible for providing access to external networks, such as the Internet, and it also provides some security and QoS features.

- Application Layer: These are the networks that are connected to the LTE network, such as the Internet or other networks. They provide access to external resources and services for the devices in the LTE network. Here, the usual TCP/IP protocols are used for communication.

Once installed, a cellular network enables connection to the Internet everywhere in the area covered by the network (which is usually really large). However, setting up a cellular network can be really expensive, and it may not be feasible for some IoT applications that require low-cost communication solutions. To address this issue, the NB-IoT standard was developed to provide a low-cost, low-power, wide-area communication solution for IoT applications that can be deployed in existing LTE networks without the need for changes to the infrastructure.

3.2.2. NB-IoT Channels

NB-IoT can be deployed in three different ways, which use different channels for communication:

- Standalone Deployment: This deployment is used when there is no other system available and therefore GSM (which is an older cellular communication standard) spectrum is used for communication. In this deployment, NB-IoT reserves a specific portion of the GSM spectrum for communication (a single 200 kHz channel), and it uses that channel for communication between the devices and the base stations.

  As an advantage, this deployment does not interfere with the existing LTE network, as it uses a different spectrum for communication. However, GSM is older and less efficient than LTE, so this deployment may not provide the best performance for NB-IoT applications.

- Guard Band Deployment: This deployment is used when there is an existing LTE network, and it uses the guard band (which is a portion of the spectrum that is reserved for preventing interference between different channels) for communication. This band is not used for communication in the LTE network, so it can be used for NB-IoT communication without interfering with the existing LTE network. A channel is then reserved in the guard band for NB-IoT communication.

- In-Band Deployment: This deployment is used when there is an existing LTE network, and it uses a portion of the LTE spectrum for communication. In this deployment, a specific portion of the LTE spectrum (a single channel) is used for NB-IoT communication, and it is time-shared with the existing LTE network. Collisions between NB-IoT and LTE communication need to be controlled to ensure that both types of communication can coexist without interference.

However, these physical channels are not directly used, but they are abstracted by logical channels that provide specific functionalities and features for enabling communication in NB-IoT applications. NB-IoT uses specific logic channels, and each of them is divided in two different types of channels:

- Signaling Channels: These channels are used for transmitting control information and signaling messages between the devices and the base stations. They are used for tasks such as device registration, authentication, and mobility management.
- Data Channels: These channels are used for transmitting user data between the devices and the base stations.

Each of these channels in fact uses several physical channels for communication, and they have to be mapped. There are two different mappings that are applied:

- Frequency Multiplexing: Two different physical channels are used, one for uplink communication (from the device to the base station) and another for downlink communication (from the base station to the device).
- Time Multiplexing: In the same physical channel, different time slots are used for different logical channels, so the devices and the base stations need to be synchronized to ensure that they are using the correct time slots for communication.

3.2.3. Cell Controlling in NB-IoT

NB-IoT has a well-known common model: messages will be infrequent and small. This allows for no handover. While you are being served by a base station, you can perfetly communicate with it, but if you move to another cell, the communication will be lost and you will need to join to the new cell. In order to join a cell, the following process is followed:

1. The UE listens in a broadcast channel for a specific message from the base station containing the information about the cell.
2. After receiving that message, the UE checks if the cell is joinable (as there are test cells, for example, that are not joinable).
3. If the cell is joinable, in the information received there is a threshold for the signal strength, so the UE checks if the signal strength is above that threshold.
4. From all the joinable cells with a signal strength above the threshold, the UE selects the one with the best signal strength and tries to join it.

Switching between cells is also supported but, as previously mentioned, there is no handover, so the communication will be lost during the switching process. The switching process can be performed when, due to mobility or other factors, the signal strength of the current cell drops below a certain threshold, and a new cell with a better signal strength is available. In that case, the UE can switch to the new cell by following the same process as when joining a cell for the first time.

3.2.4. UE Modes in NB-IoT

NB-IoT defines four different modes for the UE, which are designed to optimize the power consumption of the devices in the NB-IoT network.

Active Mode

An UE only enters this mode when it needs to transmit or receive data, and it is the most power-consuming mode. In this mode, the UE is actively communicating with the base station, and therefore has reserved resources for communication. There are two different types of transmission in this mode:

- Uplink Transmission: In this type of transmission, the UE transmits data to the base station. There are two different variants:
  - *Via Data Channels*: This is a connection-oriented transmission. The UE uses a signal channel to request resources for uplink transmission, and the base station informs the UE about the timeslots and the data channels that it can use for the uplink transmission. The UE can then transmit its data to the base station using those resources when needed. After the transmission is completed, the UE can release the resources that it has been using for the uplink transmission and close the connection.

    This is useful for large transmissions, but introduces a lot of overhead for small transmissions. An small optimization is used by letting the UE ask the base station to resume the connection that it had previously established, so the UE can use the same resources that it had been using before without the need to go through the whole process of requesting resources and establishing a new connection. However, the overhead is still significant for small transmissions, so there is another variant for small transmissions.
  - *Via Signal Channels*: This is a connectionless transmission. In the first step, when the UE asks the base station for resources for uplink transmission, the actual data that the UE wants to transmit is included in the request (called piggybacking), so there is no need to establish a connection and reserve resources for the transmission. The base station can then transmit the data to the core network immediately after receiving the request from the UE, without the need to wait for the UE to transmit its data using the reserved resources. This is useful for small transmissions, as it allows for a more efficient use of the resources and reduces the overhead associated with establishing a connection and reserving resources for the transmission.
- Downlink Transmission: In this type of transmission, the base station transmits data to the UE. The base station pages the UE (it can page several UE in the same message) to notify it that there is data waiting for it to be transmitted. Once the UE receives the paging message, there are also the two same variants for the downlink transmission:
  - *Via Data Channels*: Connection-oriented transmission where the UE reserves a specific timeslot and data channel for the downlink transmission.
  - *Via Signal Channels*: Connectionless transmission where the UE tells the base station that it is ready to receive the data, and that the data should be included (piggybacked) in the message sent over the signal channel, so there is no need to reserve resources for the downlink transmission.

Regarding the downlink transmission, paging is used to notify the UE that there is data waiting for it to be transmitted. However, once the gateways receive the data for the UE, it is not yet known which base station is serving the UE, i.e., which base station should send the paging message to the UE. To solve this issue, tracking is used. Two extreme scenarios could be considered to better understand how it is done.

- *Only one cell is paged*: The MME keeps track of the cell that is serving the UE at every moment, so when there is data waiting for the UE, only its specific cell is paged. This would reduce the overhead of paging (as only one cell is paged), but it would imply a lot of overhead each time the UE switches between cells. This is also a problem from the privacy point of view, as the MME would have to keep track of the cell that is serving the UE at every moment, which could be used to track the location of the UE.

- *All the cells are paged*: The MME does not keep track of the cell that is serving the UE, so when there is data waiting for the UE, all the cells are paged. This would reduce the overhead of tracking (as there is no need to keep track of the cell that is serving the UE), but it would imply a lot of overhead for paging (as all the cells are paged).

As the reader may have already guessed, the actual solution is a trade-off between these two extreme scenarios. The concept of *Tracking Area* is used, which is just a group of cells with the same tracking area identifier (TAI). The MME keeps track of the tracking area that is serving the UE, so when there is data waiting for the UE, only the cells in that tracking area are paged. When the UE needs to inform the MME of the cell that is serving it, it sends a Tracking Area Update (TAU) message to the MME, which updates in its internal records the tracking area that is serving the UE.

- When a UE joins the network, it sends a TAU message to the MME to inform it of the tracking area that is serving it.

- When a UE switches between cells, it checks if the new cell belongs to the same tracking area as the previous cell. If it does, there is no need to send a TAU message to the MME, as the tracking area that is serving the UE has not changed. If it does not, the UE sends a TAU message to the MME to inform it of the new tracking area that is serving it.

- Every certain period of time, the UE sends a TAU message to the MME to inform it that it is still connected to the network. If the MME does not receive a TAU message from the UE within a certain period of time, it assumes that the UE has disconnected from the network and it deletes its internal record accordingly.

Idle Mode

This mode is used when the UE does not need to transmit or receive data, but it still needs to be registered in the network and able to receive paging messages from the base station. No resources are reserved for communication in this mode, but:

- The UE still informs the base station about its location, so the base station can send paging messages to the UE when needed.
- The UE still monitors for paging messages from the base station, so it can receive them when they are sent.
- The SGW retains the incoming data for the UE in buffers until it receives a paging message from the base station, which indicates that there is data waiting for the UE to be transmitted or received.

Even though a UE can set itself to idle mode, the base station can also set the UE to idle mode if it detects that the UE has not been transmitting or receiving data for a certain period of time.

Extended Discontinuous Reception (eDRX) Mode

One thing that should be taken into account is that paging is not done continuously, but it is done in specific time intervals are the start of each hyperframe (which is a group of slotframes). This means that the UE needs to be listening for paging messages only at the start of each hyperframe, and it can go to sleep during the rest of the time to save energy.

The eDRX mode is used when the UE needs to save energy by sleeping for longer periods of time, but it still needs to be able to receive paging messages from the base station. In this mode, the UE can configure the number of hyperframes that it will sleep for, so it will only listen for paging messages at the start of each hyperframe after sleeping for that number of hyperframes. This allows the UE to save energy by sleeping for longer periods of time, while still being able to receive paging messages from the base station when needed. During all that time, the SGW retains the incoming data for the UE in buffers.

Power Saving Mode (PSM)

This mode is used when the UE needs to save even more energy by sleeping for an unlimited period of time. During this time, the base station will not page the UE, and the SGW will retain the incoming data for the UE in buffers. When the UE awakes from this mode to send data, it will notify the base station that it is awake and wait for the base station to send the buffered data to it (if there is any data waiting for it to be transmitted or received).

3.3. 6LowPAN

3.4. MQTT

4. Question Sheets

4.1. Introduction

The recorded lecture is provided here.

Ejercicio 4.1.1. What are the two views on IoT discussed in the lecture? How do they differ?

The two views on IoT are:

- IoT as a network of connected *things*, aka *M2M (Machine to Machine)*.
- IoT as a network of connected *data*, aka *Big Data*.

They differ in plenty of different aspects:

- While the first one focuses on the devices and the communication between them, the second one focuses on the data that is generated by these things and how it can be used to create value.
- While the first one sees Internet as just a communication medium, the second one sees Internet as a platform for data collection and analysis.
- While the main goal of the first one is to connect as many things as possible, the main goal of the second one is to extract insights from the data.
- While the first one is usually taken by engineers and computer scientists, the second one is usually taken by business people and data scientists.

Ejercicio 4.1.2. Are IoT gateways always necessary? What are they used for?

IoT gateways are used to connect IoT things to the internet. IoT things are usually resource-constrained devices that cannot connect directly to the internet, so they need a gateway to act as a bridge between them and the internet. However, IoT gateways are not always necessary, as some IoT things can connect directly to the internet using low-power communication technologies such as Wi-Fi or cellular. In this case, the IoT thing itself can act as a gateway, and there is no need for an additional device.

4.2. IoT Systems

The recorded lecture is provided here.

Ejercicio 4.2.1. What is the difference between an IoT system and an IoT platform? Use the WWW to find another IoT platform that is not mentioned in the lecture.

They are different concepts. An IoT system consists of all the HW and SW components that are required to implement an IoT application, and developpers tend to reuse the same components across different applications. This is where the concept of IoT platforms comes into play, because an IoT platform is a framework that provides a set of tools and services to facilitate the development, deployment, and management of IoT applications. By using an IoT platform, developers can focus on the application logic rather than dealing with the complexities of the underlying infrastructure. Therefore, an IoT platform is a tool that can be used to build an IoT system.

An example of an IoT platform that is not mentioned in the lecture is Blynk. It is an horizontal IoT platform that provides a set of tools and services to facilitate the development, deployment, and management of IoT applications. It allows developers to create custom IoT applications using a drag-and-drop interface, and it supports a wide range of hardware platforms and communication protocols.

Ejercicio 4.2.2. Is the Microsoft Azure IoT Suite a horizontal or a vertical IoT platform? Why?

It is an horizontal IoT platform because it provides a broad set of features and services that can be customized to fit the specific needs of different applications. It is not designed for a specific industry or use case, but rather it can be used for a wide range of IoT applications across different domains.

Ejercicio 4.2.3. Provide an example (i.e., search on the Internet or come up with your own idea) for a scenario in which you would use predictive data processing but not prescriptive data processing. What is the main legal difference between these?

There are plenty of cases where predictive data processing can be used without prescriptive data processing for several reasons:

- Prescriptive data processing can simply not be made.

  For instance, an IoT application designed to predict the weather conditions in a specific area based on historical data and patterns. The application can provide valuable insights and information about the expected weather conditions, but the weather cannot be changed by the application, so prescriptive data processing is not possible in this case.

- Prescriptive data processing can be made, but it is not ethical to do it without human supervision and intervention.

  For instance, an IoT application designed to predict the likelihood of a patient developing a certain medical condition based on their health data and lifestyle habits. The application may recommend certain lifestyle changes or medical interventions to reduce the risk of developing the condition, but it is not ethical to implement these recommendations without human supervision.

- Prescriptive data processing can be made, but a responsible person must be legally accountable for the consequences of the actions taken based on the prescriptive data processing.

  For instance, an IoT application designed to predict the future of the stock market based on historical data and patterns. The application may recommend certain investment strategies to maximize returns, but someone should be legally accountable for the consequences of the actions taken based on these recommendations, as the stock market is highly volatile and unpredictable, and there is a risk of financial loss.

The main legal difference between predictive and prescriptive data processing is that prescriptive data processing actually performs actions or makes decisions, and they can have legal consequences, so there must be a responsible person who is legally accountable for those consequences.

Ejercicio 4.2.4. What are the differences between mesh-based and edge-based IoT systems? Which one is currently supported by platform providers?

Both IoT system architectures are designed to process data closer to where it is generated, but in the edge-based architecture the IoT things are still really simple and resource-constrained, so they just collect data and send it to the gateway, which is responsible for processing the data and making decisions based on it. In contrast, in the mesh-based architecture, the IoT things are more powerful and can communicate directly with each other, so they can process data and make decisions without the need for a central gateway.

Currently, even though the most supported architecture by platform providers is the cloud-based one, there are some platforms that support edge-based IoT systems, but mesh-based IoT systems are still not widely supported by platform providers due to the complexity and challenges associated with implementing and managing such systems.

4.3. IoT Internetworking

4.3.1. Motivation

The recorded lecture is provided here.

Ejercicio 4.3.1. Why do we need new protocols for the IoT instead of using established Internet protocols? Describe one challenge existing protocols face.

New protocols are needed for the IoT because existing Internet protocols, such as TCP/IP, are not optimized for the resource-constrained nature of IoT devices. One challenge existing protocols face is that they may require larger buffer sizes than what is available on IoT devices, which can lead to performance issues and increased latency.

Ejercicio 4.3.2. Which central protocol can we not replace (completely) for the IoT? Why is it harder to replace it than others?

The central protocol that we cannot replace (completely) for the IoT is the Internet Protocol (IP). It is harder to replace IP than other protocols because it is the fundamental protocol that enables communication and data exchange between different devices and applications on the Internet. It is the backbone of the Internet and, technically speaking, being connected to the Internet requires having an IP address and using IP for communication. Therefore, while we can use different protocols for the application layer or the transport layer, we still need to use IP at the network layer in order to enable connectivity and communication with the Internet. In addition, replacing IP would require a complete overhaul of the Internet infrastructure and would require all devices and applications to adopt the new protocol, which is a significant challenge in terms of compatibility and interoperability.

Ejercicio 4.3.3. We introduced two approaches for Smart Things to connect to the Internet. Name these approaches, describe them briefly and give one advantage and one disadvantage of each, compared to the other.

The two approaches for Smart Things to connect to the Internet are:

- Application layer proxy: In this approach, there is a proxy (usually implemented in the IoT gateway) that translates between the protocols used by the IoT Things and the protocols used by the Internet. The IoT Thing communicates with the proxy using dedicated IoT protocols that have nothing to do with the Internet protocols, and the proxy stores the data received from the IoT Things and makes it available to the Internet using standard Internet protocols.
    - Advantage: It allows for the use of dedicated IoT protocols that are optimized for the resource-constrained nature of IoT devices, which can lead to better performance and energy efficiency.
    - Disadvantage: It introduces an additional layer of complexity and can lead to increased latency and reduced scalability (due to the dedicated software), as all communication between the IoT Things and the Internet needs to go through the proxy.

- IP Router: In this approach, the IoT Things directly use IP, but with some optimizations and adaptations to address the challenges and limitations mentioned above. There is no need for specific apps or software to interact with the IoT Things, as they can be accessed using standard Internet protocols and interfaces.
  - Advantage: It provides better scalability and interoperability, as the IoT Things can communicate directly with other devices and applications on the Internet without the need for a proxy.
  - Disadvantage: It may require more complex and resource-intensive protocols and technologies to enable efficient communication and data exchange between the IoT Things and the Internet, which can lead to increased power consumption and reduced performance for resource-constrained IoT devices.

4.3.2. IEEE 802.15.4

The recorded lectures are provided here (Part A) and here (Part B).

Ejercicio 4.3.4. What are the differences between the PHY and the MAC layer?

Both are layers in the OSI model, but they serve different purposes. The PHY (Physical) layer is responsible for the physical transmission of data over the communication medium. The MAC (Medium Access Control) layer builds on top of the PHY layer and is responsible for managing access to the communication medium, including addressing, framing, and error detection.

Ejercicio 4.3.5. Discuss pros and cons of using 16 bit addresses for 802.15.4 instead of 64 bit addresses.

Some pros of using 16 bit addresses for 802.15.4 include:

- Reduced overhead: 16 bit addresses require less memory and bandwidth compared to 64 bit addresses.
- Faster processing: 16 bit addresses can be processed more quickly than 64 bit addresses, which can lead to improved performance and reduced latency.

Some cons of using 16 bit addresses for 802.15.4 include:

- Limited address space: 16 bit addresses can only support up to $2^{16} = 65536$ unique addresses, which may not be sufficient for large-scale IoT deployments.
- Not globally unique: 16 bit addresses are not globally unique, so they can only be used within a local network, which can limit interoperability and scalability.
- Harder to manage: They have to be managed and assigned manually, which can lead to errors and conflicts in larger networks.
- If a device moves to another network, it may need to change its address, which can lead to additional overhead and complexity in managing the network.

Ejercicio 4.3.6. Describe the basic idea of CSMA/CA and compare it to frequency hopping. You may use a diagram / drawing to describe your answer.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) is a protocol used in wireless communication to manage access to the communication medium. It works by having devices listen to the channel before transmitting data. If the channel is busy, the device will wait for a random backoff time before trying again. This helps to avoid collisions and improve the efficiency of the network. In contrast, frequency hopping is a technique where devices rapidly switch between different frequency channels to avoid interference and improve security. The hopping pattern is typically determined by a pseudo-random sequence, so no two devices will hop to the same frequency at the same time, which means that they can transmit simultaneously without interference.

Ejercicio 4.3.7. Briefly describe and compare TDMA and FDMA. Which one is better?

TDMA (Time Division Multiple Access) is a protocol that divides the communication medium into time slots, and each device is assigned the whole frequency channel for a specific time slot. This means that, while only one device can transmit at a time, it can use the full bandwidth of the channel during its assigned time slot. FDMA (Frequency Division Multiple Access) is a protocol that divides the communication medium into frequency channels, and each device is assigned a specific frequency channel for transmission. This means that multiple devices can transmit simultaneously, but they have to share the bandwidth of the channel. The choice between TDMA and FDMA depends on the specific requirements of the application and the characteristics of the communication medium.

- Depending on the application.

  TDMA may be better for applications that require high data rates but can tolerate some delay, while FDMA may be better for applications that require no delay but can tolerate lower data rates.

- Depending on the channel characteristics.

  If the bandwidth of the channel is limited, TDMA may be more efficient as it allows devices to use the full bandwidth during their assigned time slots. On the other hand, if the channel has a large bandwidth and can support multiple simultaneous transmissions, FDMA may be more efficient as it allows multiple devices to transmit at the same time without interference.

- Depending on the number of devices.

  If there are a large number of devices that need to access the communication medium, FDMA may be more efficient as it allows multiple devices to transmit simultaneously, while TDMA may lead to increased latency and reduced performance due to the need for devices to wait for their assigned time slots.

Ejercicio 4.3.8. An 802.15.4 network is deployed in the following three application scenarios. Which CSMA/CA MAC Layer version would you use? Justify your answer (e.g., “In this case, we should use slotted CSMA/CA, because...”).

- A smart home network, where only 10 temperature sensors are installed. Each sensor reports a measurement once per minute.

  Given the low number of devices, the low data rate, and the fact that the devices are not synchronized, we can use unslotted CSMA/CA, as it is simpler and more efficient for this scenario.

- A smart clinic network that transmits monitoring data about the patients and alarm notifications.

  In this case, we should use slotted CSMA/CA, because it can provide better performance and reduced latency for time-sensitive applications, such as monitoring data. In addition, thanks to the GTS mechanism, it can provide guaranteed time slots for alarm notifications, which can be critical for patient safety.

Ejercicio 4.3.9. When unslotted CSMA/CA is applied, at each trial the back-off window size will be doubled. However, it doesn't always lead to a longer delay, why?

The back-off window size is doubled after each failed transmission attempt, but this does not always lead to a longer delay because the back-off time is randomly selected from the back-off window. The expected back-off time may be doubled, but it is possible that the randomly selected back-off time is shorter than the previous one, which can lead to a shorter delay.

4.3.3. NB-IoT

The recorded lectures are provided here (Part A) and here (Part B).

Ejercicio 4.3.10. Explain the basic architecture of a cellular network. How does it cover large areas?

A cellular network is a wireless communication network that is divided into smaller geographic areas called cells. Each cell is served by a base station, which is responsible for providing wireless coverage and connectivity to the devices within its cell. The base stations are connected to a central network infrastructure, which manages the communication and data exchange between the devices and the Internet. The cellular network covers large areas by using multiple cells that are strategically placed to provide coverage to the desired geographic area. The cells can be of different sizes, depending on the population density and the demand for wireless services in the area. By using multiple cells, the cellular network can provide coverage to a large number of devices while maintaining good performance and reliability.

Ejercicio 4.3.11. Briefly describe an application case in which NB-IoT is better than IEEE 802.15.4. Justify your answer.

An application case in which NB-IoT is better than IEEE 802.15.4 is whenever you need to cover a large area with a low density of devices, such as in smart agriculture or smart city applications. NB-IoT is designed for wide-area coverage and can support a large number of devices over a wide geographic area, while IEEE 802.15.4 is designed for short-range communication and is better suited for applications that require low power consumption and low data rates. However, IEEE 802.15.4 should not be forgotten, as it works in the unlicensed spectrum and can be used without the need for a cellular network infrastructure, which can be advantageous in certain scenarios, such as in remote areas or in applications that require low-cost deployment.

Ejercicio 4.3.12. Name and briefly explain two of the possible deployment scenarios that specify the frequency range that NB-IoT may be deployed in.

Two possible scenarios for NB-IoT deployment are:

- In-band deployment: In this scenario, NB-IoT is deployed within the existing LTE spectrum, using the same frequency bands as LTE. This allows for easy integration with existing LTE infrastructure and can provide good performance and coverage for NB-IoT devices. However, it may lead to increased interference and reduced performance for both NB-IoT and LTE devices, as they are sharing the same frequency bands.

- Guard-band deployment: In this scenario, NB-IoT is deployed in the guard bands between LTE frequency bands. This allows for better isolation and reduced interference between NB-IoT and LTE devices, as they are using different frequency bands. However, it may require additional spectrum allocation and may not be as widely supported as in-band deployment, which can limit the availability and coverage of NB-IoT devices.

Ejercicio 4.3.13. How are frequency and time multiplexing used in NB-IoT?

NB-IoT uses both frequency and time multiplexing to enable efficient communication and data exchange between the devices and the base station.

- Frequency multiplexing: NB-IoT uses frequency division multiplexing (FDM) to divide the available frequency spectrum into two different types of channels: uplink channels for transmitting data from the devices to the base station, and downlink channels for transmitting data from the base station to the devices. This allows multiple devices to transmit and receive data simultaneously without interference, as they are using different frequency channels.

- Time multiplexing: NB-IoT uses time division multiplexing (TDM) to divide the physical channel into time slots, and each time slot is assigned to a specific logical channel (e.g., control channel, data channel, etc.). This allows multiple devices to share the same frequency channel by transmitting and receiving data in different time slots.

Ejercicio 4.3.14. Explain the procedure that a newly powered on user equipment (UE) follows to join a cell.

When a newly powered on user equipment (UE) wants to join a cell, it follows the following procedure:

- The UE listens in a broadcast channel for a signal from base stations, which contains information about the available cell and its characteristics.

- The UE checks if the cell is joinable, as some cells may be reserved for specific types of devices or applications.

- If the cell is joinable, the UE checks if it has enough signal strength to connect to the cell.
- Among the cells that are joinable and have enough signal strength, the UE selects the one with the best signal quality (e.g., the one with the highest signal-to-noise ratio) and sends a connection request to the base station of that cell.

Ejercicio 4.3.15. What is a tracking area in NB-IoT and how is it used?

A tracking area in NB-IoT is a group of cells that share the same tracking area identifier (TAI). The MME (Mobility Management Entity) keeps track of the tracking area that is serving the UE, so when there is data waiting for the UE, only the cells in that tracking area are paged. When the UE needs to inform the MME of the cell that is serving it, it sends a Tracking Area Update (TAU) message to the MME, which updates in its internal records the tracking area that is serving the UE. This allows for a trade-off between the overhead of paging and the overhead of tracking, as it reduces the overhead of paging while still allowing for efficient tracking of the UE's location within the network.

Ejercicio 4.3.16. How can a UE use Extended Discontinuous Reception to save energy?

Extended Discontinuous Reception (eDRX) is a mode that allows the UE to save energy by sleeping not just between the start of each hyperframe (when paging is performed), but for a configurable number of hyperframes. This means that the UE can configure the number of hyperframes that it will sleep for, so it will only listen for paging messages at the start of each hyperframe after sleeping for that number of hyperframes. This allows the UE to save energy by sleeping for longer periods of time, while still being able to receive paging messages from the base station when needed. During all that time, the SGW retains the incoming data for the UE in buffers.